

EDUCATION

University of Toronto

HBSc, Specialist in Data Science and Major in Computer Science

- CGPA: 3.79, **Dean's List** 2024 and 2025

Toronto, Ontario

Expected June 2026

PROJECTS

Salt, Grooves, and Garlic (itch.io)

Sep 2025 - Present

Unity, C#

Toronto, Ontario

- **2nd Best Game Overall** at **LevelUp 2026**, rhythm-action game blending **osu!/Cytus** timing with **Crypt of the NecroDancer** grid-based movement, developed for **CSC404** game design capstone with a 10-person cross-functional team and a public demo on itch.io
- Architected core **game manager** and centralized state control, coordinating scene flow, beat synchronization, and gameplay subsystems across the full game loop
- Designed **extensible boss state machine** with a modular base attack pattern system, streamlining new attack development and accelerating gameplay iteration
- Built **data-driven sequence runner** for tutorial flow, executing ordered **ScriptableObject** events that designers can add, configure, and reorder in the Unity Editor with hot-reload
- Engineered **dynamic audio manager** with seamless music cross-fading during tutorial-to-gameplay transitions, eliminating audible gaps and preserving rhythm immersion
- Owned end-to-end implementation of the **tutorial level**, including event scripting, dialogue triggers, music transitions, and player guidance
- Implemented **dialogue system** for narrative delivery and tutorial guidance, supporting designer-authored content via **ScriptableObject** data
- Built **pause/resume** with full game-state preservation, correctly handling rhythm timing, audio, and active **coroutines** without desync
- Redesigned and implemented **main menu** layout, integrating new 3D assets from the art team to elevate first-impression presentation
- Built contextual **UI overlay** that dynamically shows and hides input prompts based on nearby interactable objects, improving onboarding discoverability
- Drove **gameplay polish** pass across menus, transitions, and UI feedback for the LevelUp showcase
- Integrated Unity's **New Input System** with rebindable action maps for keyboard and gamepad support

Ant Colony Simulation ([github](https://github.com))

Jan 2026

C++, Ubisoft NEXT API

Ubisoft NEXT

- Solo submission to **Ubisoft NEXT 2025-2026** ("Simulation of Large Number of Entities" challenge), simulating two rival ant colonies with pheromone pathfinding and emergent swarm behaviors at **2000 entities** in real-time
- Built end-to-end in **C++** with only the framework's primitive draw calls (line, triangle, rectangle), learning **ECS** from scratch over 3-4 weeks before 3 days of focused implementation
- Designed cache-friendly **ECS** with **Structure of Arrays (SoA)** memory layout (`std::tuple<std::vector<T>...>`) for contiguous component storage and CPU cache efficiency
- Implemented **O(1) component lookups** via per-entity bitmasks and bitwise AND, eliminating hash map overhead in the hot path
- Built per-type **cached entity lists**, rebuilt on add/remove instead of filtering 10,000 entities each frame
- Architected modular **system pipeline** (Ant, Spider, Collision, Drag, Movement, Render) with a deterministic frame update order
- Engineered **uniform spatial grid** (32-unit cells) with templated **QueryEach** lambda interface for **zero-allocation** neighbor iteration
- Optimized AI proximity logic with combined **NearbyQuery::QueryAll**, returning nearest spider, enemy ant, and food in one grid pass instead of three
- Designed **5-state FSM** for ant AI (WANDER, FOLLOW_TRAIL, FORAGE, ATTACK, FLEE) with per-frame **AntContext** bundling refs and queries to eliminate redundant lookups
- Implemented **3-layer pheromone system** (HOME per-team, FOOD shared, ALARM combat) producing emergent foraging trails, combat coordination, and team-specific homing
- Built **Monte Carlo cone sampling** for pheromone pathfinding (5 forward samples vs 25-neighbor gradient), preventing traffic jams and producing natural path variety
- Engineered **cooperative dragging** for up to 8 ants per food item, sampling pheromones once at the food's position to keep dragger directions aligned
- Built **stuck detection** via rolling position history, forcing escape vectors when ants fail to make progress in dense crowds
- Pre-computed **sin/cos lookup tables** (32 entries) at startup, eliminating runtime trig in the AI hot loop
- Implemented **staggered pheromone decay** with per-layer tick timers, distributing decay cost across frames instead of spiking each update
- Applied **frustum culling** via camera visibility checks before draw submission, skipping off-screen entities entirely

- Engineered **screen-space LOD** selecting circle segment count (4, 6, 8, 12) by radius times zoom, rendering distant ants as cheap diamonds and close-ups as smooth circles
- Designed **typed event buffer** for collisions (AntFood, AntSpider, FoodColony), separating detection from response across two passes to avoid modification-during-iteration bugs
- Built **Observer-pattern action system** mapping hardware input to named actions, letting scenes subscribe to actions instead of raw keycodes
- Implemented **debug visualization** with runtime toggles for HOME, FOOD, and ALARM pheromone layers, supporting iterative AI tuning
- Built **2D camera** with world-to-screen transforms, pan, and zoom on top of the framework's primitive draw calls

EXPERIENCE

Full-Stack Developer → Tech Lead

Aug 2025 - Present

Ontario Association of Physics Teachers (OAPT), University of Toronto

Toronto, Ontario

- Promoted to **Tech Lead** after 5 months as Full-Stack Developer, owning end-to-end architecture and delivery of a contest platform designed for **660,000+ students** across **883+ Ontario schools**
- Architected full-stack web application using **React**, **Node.js/Express.js**, **PostgreSQL**, and **Prisma ORM**, designing a normalized database schema and **RESTful API** with **20+** endpoints
- Built secure authentication with **JWT tokens**, **bcrypt** password hashing, and **role-based access control (RBAC)** supporting student, teacher, and admin roles
- Built rich-text contest editor using **TipTap/ProseMirror** with **LaTeX** math rendering, image uploads via **Multer**, drag-and-drop question ordering (**@dnd-kit**), and **PDF generation** via **Puppeteer**
- Implemented responsive React UI with **Material-UI** centralized theming, custom hooks for route protection, and **Context API** for global state management
- Used **PostgreSQL transactions** to ensure data integrity across multi-step operations, preventing partial writes and maintaining database consistency
- Lead a team of **7 developers** alongside a senior developer using **Agile** principles, conducting GitHub PR code reviews and mentoring on Git workflows, RESTful API design, and React patterns
- Established team-wide development standards from scratch, including **PR templates**, issue tagging conventions, branching strategy, and code review expectations
- Restructured team workflows to distribute issue creation across all developers and built a **promotion path** for contributors to advance into PR reviewer roles
- Built a **Discord webhook integration** pushing real-time notifications for issue and PR updates with manual developer tagging, eliminating missed communications and replacing unreliable email-based updates
- Responded to a live security incident caused by a compromised npm package in the dependency chain, coordinating containment and implementing improved **dependency auditing** practices afterward

SKILLS

Programming Languages: C++, C#, JavaScript/TypeScript, Python, Java, R, C, MIPS Assembly

Game Engines & Frameworks: Unity, Ubisoft NEXT, Unreal Engine (learning)

Game Programming: ECS, Gameplay Systems, AI/FSM, Spatial Partitioning, Game Loop Architecture, Component-Based Design, Data-Driven Design, ScriptableObjects, Coroutines, Object Pooling

Graphics & Math: 2D Camera Systems, Frustum Culling, Level of Detail (LOD), Vector Math, Blinn-Phong Shading

Performance & Optimization: Cache-Friendly Memory Layout, Structure of Arrays (SoA), Lookup Tables, Zero-Allocation Patterns

Tools & Workflow: Git/GitHub, Visual Studio, Unity Editor, Trello, Slack, Linux/Unix, Shell Scripting

Web Stack: React, Next.js, Node.js, Express.js, PostgreSQL, Prisma ORM, RESTful APIs, JWT, RBAC, Material-UI, Tailwind CSS, Docker, AWS

Specialized Libraries: TipTap/ProseMirror, Puppeteer, Multer, LaTeX

AWARDS & ACTIVITIES

Awards: Indonesia Maju Scholarship

Teaching: Teaching Assistant for CSC309 (Programming on the Web), University of Toronto

Leadership: Led Recognized Study Group for CSC165 (Mathematical Expression and Reasoning for CS)

Languages: Indonesian (native), English (fluent)